Ex.No.1	INTRODUCTION TO DATA SCIENCE PYTHON PROGRAMMING	URK24CS1273
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Aim:

Write a python program to demonstrate the basics of dictionary, list, set, tuple, array and csv files for data science application.

Procedure:

Python is a high-level general-purpose programming language. Python is one of the most popular and widely used programming languages in the world Python is used for data analytics, machine learning, and even design. Python is used in a wide range of domains owing to its simple yet powerful nature. Python provide great functionality to deal with mathematics, statistics and scientific function. It provides great libraries to deals with data science application. One of the main reasons why Python is widely used in the scientific and research communities is because of its ease of use and simple syntax which makes it easy to adapt for people. It is also more suited for quick prototyping. Hence this language is mainly used for Data Science applications.

Question 1:

Create an empty dictionary and fill with some book_id and book_name as pair by user input. Then take one book_id as input from the user and traverse through dictionary to find the corresponding book_name and display the same.

Program:

```
b = {}
for _ in range(int(input("Enter number of books: "))):
    b[input("Enter Book ID: ")] = input("Enter Book Name: ")
search = input("\nEnter Book ID to search: ")
print(f"Book Name: {b.get(search, 'Not Found')}")
```

Output Screenshot:

```
Enter number of books:  2
Enter Book Name:  1
Enter Book ID:  sql
Enter Book Name:  2
Enter Book ID:  data base management

Enter Book ID to search:  3
Book Name: Not Found
```

Question 2:

Create an empty dictionary and fill with some sub_code and sub_name as pair by user input. Then take one sub_code as input from the user and traverse through dictionary to find the corresponding sub_name and display the same.

Program:

```
subjects = {}
for _ in range(int(input("Enter number of subjects: "))):
    subjects[input("Enter Subject Code: ")] = input("Enter Subject Name: ")
search = input("\nEnter Subject Code to search: ")
print(f"Subject Name: {subjects.get(search, 'Not Found')}")
```

Output Screenshot:

```
Enter number of subjects:  3
Enter Subject Name:  data science ecosystem
Enter Subject Code:  23cs2025
Enter Subject Name:  web technology
Enter Subject Code:  23cs2034
Enter Subject Name:  theory of compuatation
Enter Subject Code:  22cs2002

Enter Subject Code to search:  23cs2025
Subject Name: data science ecosystem
```

Question 3:

Create an empty list and fill with list of strings by user input. Find the number of strings where the string length is 2 or more and the first and last character are same.

Program:

```
strings_list = []
for _ in range(int(input("Enter number of strings: "))):
    strings_list.append(input("Enter string: "))
count = sum(1 for s in strings_list if len(s) >= 2 and s[0] == s[-1])
print(f"Expected Result : {count}")
```

Output Screenshot:

```
Enter number of strings:  4
Enter string:  abc
Enter string:  xyz
Enter string:  aba
Enter string:  1221
Expected Result : 2
```

Question 4:

Create an empty list and fill with list of words by user input. Find the list of words that are longer than n from a given list of words.

Program:

```
words_list = []
for _ in range(int(input("Enter number of words: "))):
    words_list.append(input("Enter word: "))
n = int(input("Enter value of n: "))
result = [word for word in words_list if len(word) > n]
print(f"Expected Result : {' '.join(result)}")
```

Output Screenshot:

```
Enter number of words:  4
Enter word:  apple
Enter word:  ace
Enter word:  manga
Enter word:  nii
Enter value of n:  3
Expected Result : apple manga
```

Question 5:

Create an empty set and fill with some values by user input. Find maximum value in a set.

Program:

```
total = int(input("Total numbers: "))
numbers = []
for _ in range(total):
    num = int(input("Enter number: "))
    numbers.append(num)
print("Maximum value:", max(numbers)
```

Output Screenshot:

```
Enter number of elements:  3
Enter value:  734
Enter value:  596
Enter value:  659
Maximum Value = 734
```

Question 6:

Create an empty set and fill with some values by user input. Check if a given value is present in a set or not.

Program:

```
total = int(input("Total values: "))
values = []
for _ in range(total):
    item = input("Enter value: ")
    values.append(item)
search = input("\nEnter value to search: ")
if search in values:
    print("Present: True")
else:
    print("Present: False")
```

Output Screenshot:

```
Total values:  4
Enter value:  34
Enter value:  23
Enter value:  12
Enter value:  21

Enter value to search:  38
Present: False
```

Question 7:

Create an empty tuple and fill with some values by user input. Find the sum of tuple elements.

Program:

```
total = int(input("Total numbers: "))
numbers = []
for _ in range(total):
    num = int(input("Enter number: "))
    numbers.append(num)
my_tuple = tuple(numbers)
print("Sum of elements:", sum(my_tuple))
```

Output Screenshot:

```
Enter number of elements:  3
Enter value:  45
Enter value:  8888
Enter value:  65
Sum = 8998
```

Question 8:

Create an empty tuple and fill with some values by user input. Count the occurrence of a given input number in the tuple.

Program:

```
numbers = input("Enter numbers: ")
target = input("Number to count: ")
items = tuple(numbers.split())
print("Result:", items.count(target))
```

Output Screenshot:

```
Enter numbers:  50 10 60 70 50
Number to count:  50
Result: 2
```

Question 9:

Create a 2D array and perform matrix addition using numpy.

Program:

```
import numpy as np
r = int(input("Rows: "))
c = int(input("Columns: "))
a = []
b = []
print("Enter First Matrix")
for i in range(r):
    row = []
    for j in range(c):
        row.append(int(input()))
    a.append(row)
print("Enter Second Matrix")
for i in range(r):
    row = []
    for j in range(c):
        row.append(int(input()))
    b.append(row)
a = np.array(a)
b = np.array(b)
print(a + b)
```

Output Screenshot:

```
Rows:  2
Columns:  2
Enter First Matrix
 1
 2
 3
 4
Enter Second Matrix
 5
 6
 7
 8
[[ 6  8]
 [10 12]]
```

Question 10:

Create a 2D array and perform matrix subtraction using numpy.

Program:

```
import numpy as np
r = int(input("Rows: "))
c = int(input("Columns: "))
a = []
b = []
print("Enter First Matrix")
for i in range(r):
    row = []
    for j in range(c):
        row.append(int(input()))
    a.append(row)
print("Enter Second Matrix")
for i in range(r):
    row = []
    for j in range(c):
        row.append(int(input()))
    b.append(row)
a = np.array(a)
b = np.array(b)
print(a - b)
```

Output Screenshot:

```
Rows:  2
Columns:  2
Enter First Matrix
 1
 2
 3
 4
Enter Second Matrix
 5
 6
 7
 8
[[-4 -4]
 [-4 -4]]
```

Question 11:

Read on .csv file and display the basic details.

Program:

```
import pandas as pd
df = pd.read_csv(r"C:\Users\STUDENTS\Downloads\industry.csv")
print(df.head())
print(df.shape)
```

Output Screenshot:

```
                        Industry
0             Accounting/Finance
1   Advertising/Public Relations
2             Aerospace/Aviation
3  Arts/Entertainment/Publishing
4                     Automotive
(43, 1)
```

Scenario Question - 1: Student Marks using List

A class teacher wants to store the marks of five students and perform basic operations.
Tasks
1. Create a list containing marks of 5 students.
2. Display all the marks.
3. Find the highest and lowest marks.
4. Calculate the average mark.
5. Add a new student's mark to the list.
6. Remove the lowest mark from the list..

Program:

```
marks = [85, 78, 92, 88, 90]
print("All marks:", marks)
highest = max(marks)
lowest = min(marks)
print("Highest mark:", highest)
print("Lowest mark:", lowest)
total = sum(marks)
count = len(marks)
average = total / count
print("Average mark:", average)
marks.append(95)
print("After adding new mark:", marks)
marks.remove(lowest)
print("After removing lowest mark:", marks)
```

Output Screenshot:

```
All marks: [85, 78, 92, 88, 90]
Highest mark: 92
Lowest mark: 78
Average mark: 86.6
After adding new mark: [85, 78, 92, 88, 90, 95]
After removing lowest mark: [85, 92, 88, 90, 95]
```

Scenario Question - 2: Student Profile using Dictionary

A college stores each student's details in a digital system.

Tasks

Create a dictionary with the following information:

- Register Number
- Name
- Department
- CGPA
- Email

Perform the following:

1. Display the complete dictionary.
2. Update the CGPA.
3. Add a new key called "Semester".
4. Display only the student's Name and CGPA.

Program:

```
student = {}

student["Register Number"] = input("Enter Register Number: ")
student["Name"] = input("Enter Name: ")
student["Department"] = input("Enter Department: ")
student["CGPA"] = float(input("Enter CGPA: "))
student["Email"] = input("Enter Email: ")

print("Complete profile:", student)

new_cgpa = float(input("Enter updated CGPA: "))
student["CGPA"] = new_cgpa

semester_value = input("Enter Semester value: ")
student["Semester"] = semester_value

print("Student Name:", student["Name"])
print("Student CGPA:", student["CGPA"])
```

Output Screenshot:



Scenario Question – 3 Reading a CSV File

A school maintains student information in a CSV file.
student.csv

RollNo	Name	Department	CGPA
101	Anu	CSE	8.5
102	Bala	ECE	7.9
103	Charan	IT	8.8
104	Divya	CSE	9.2
105	Eva	AIDS	8.1

Tasks

1. Read the CSV file using Pandas.
2. Display the first five rows.
3. Display only the Name column.
4. Display only the Department column.
5. Find the average CGPA.
6. Find the student with the highest CGPA.
7. Display only CSE students.

Program:

```
import pandas as pd
file_path = input("Enter name of CSV file: ")
df = pd.read_csv(file_path)
print("\nFirst 5 rows:")
print(df.head())
print("\nNames:")
print(df["Name"])
print("\nDepartments:")
print(df["Department"])
print("\nAverage CGPA:")
print(df["CGPA"].mean())
print("\nHighest CGPA student:")
print(df[df["CGPA"] == df["CGPA"].max()])
print("\nCSE Students:")
print(df[df["Department"] == "CSE"])
```

Output Screenshot:

```
Enter name of CSV file:  student.csv

First 5 rows:
   RollNo    Name Department  CGPA
0     101     Anu        CSE   8.5
1     102    Bala        ECE   7.9
2     103  Charan         IT   8.8
3     104   Divya        CSE   9.2
4     105     Eva       AIDS   8.1

Names:
0       Anu
1      Bala
2    Charan
3     Divya
4       Eva
Name: Name, dtype: object

Departments:
0     CSE
1     ECE
2      IT
3     CSE
4    AIDS
Name: Department, dtype: object

Average CGPA:
8.5

Highest CGPA student:
   RollNo   Name Department  CGPA
3     104  Divya        CSE   9.2

CSE Students:
   RollNo   Name Department  CGPA
0     101    Anu        CSE   8.5
3     104  Divya        CSE   9.2
```

Result:

Thus the python programs using List, Dictionary, Set, Tuple, NumPy Array and CSV file operations have been successfully executed.